
Towards Better Branching Policies: Leveraging the Sequential Nature of Branch-and-Bound Tree

Anonymous Author(s)

Affiliation

Address

email

Abstract

The branch-and-bound (B&B) method is a dominant exact algorithm for solving Mixed-Integer Linear Programming problems (MILPs). While recent deep learning approaches have shown promise in learning branching policies using instance-independent features, they often struggle to capture the sequential decision-making nature of B&B, particularly over long horizons with complex inter-step dependencies and intra-step variable interactions. To address these challenges, we propose Mamba-Branching, a novel learning-based branching policy that leverages the Mamba architecture for efficient long-sequence modeling, enabling effective capture of temporal dynamics across B&B steps. Additionally, we introduce a contrastive learning strategy to pre-train discriminative embeddings for candidate branching variables, significantly enhancing Mamba's performance. Experimental results demonstrate that Mamba-Branching outperforms all previous neural branching policies on real-world MILP instances and achieves superior computational efficiency compared to the advanced open-source solver SCIP. The source code can be accessed via an anonymized repository at <https://anonymous.4open.science/r/Mamba-Branching-B4B4/>.

1 Introduction

Mixed Integer Linear Programming problems (MILPs) constitute a class of computationally challenging NP-hard problems with widespread applications across diverse domains, including scheduling [7], planning [37], and transportation [3]. The branch-and-bound (B&B) method [28] represents the predominant solution methodology for MILPs in practice. This approach begins with the relaxation of the original problem and iteratively branches on variables that violate integer constraints. By maintaining global upper and lower bounds, the method progressively converges toward an optimal solution. Many high-performance MILP solvers such as SCIP [6] and Gurobi [17] employ the B&B framework as their core solution architecture.

Within the B&B framework, the selection of branching variables plays a critical role in determining computational efficiency. To this end, learning-based branching methods have been proposed [12, 15, 16, 40]: by constructing a bipartite graph that incorporates instance features and intra-tree dynamic features, and utilizing graph convolutional networks (GCNN) [12] for state encoding. Nevertheless, reliance on instance-specific features restricts their generalization to heterogeneous MILP instances. To enable cross-instance adaptability, recent approaches have focused on parameterizing the B&B tree to construct a shared feature space independent of specific problem data. For example, Zarpellon et al. [48] develop a parameterized B&B tree framework to create a shared feature space, decoupling branching decisions from instance-specific features. Further advancing this approach, T-BranT [31] evaluates the mutual connections between candidate variables by the self-attention mechanism and employs Graph Attention Networks to encode the empirical branching history in the search tree.

37 However, existing works universally overlook the sequential nature inherent in B&B tree expansion. In
38 this paper, our key insight lies in that the “branching path” from the root node to the optimal solution
39 node essentially constitutes a serialization process. This “branching path”, which encompasses
40 the parameterized tree states and corresponding branching variables from each preceding step,
41 significantly influences the current branching decision. While T-BranT incorporates historical data,
42 it models the tree from an unordered graph perspective, failing to explicitly capture this essential
43 sequential nature. Effectively modeling this sequential nature presents two key challenges: (1) Design
44 of long sequence modeling architectures. The sequence model must simultaneously capture inter-step
45 dependencies and intra-step candidate variable relationships. Given that each state comprises multiple
46 candidate variables, the length of the sequence input will increase exponentially with the number of
47 branching steps. Therefore, it is essential to develop specialized architectures that can accommodate
48 ultra-long sequences. (2) Construction of discriminative feature embeddings. An embedding layer
49 needs to be designed to map the features of candidate variables into a high-dimensional vector space
50 with high discriminative power. This will enable the sequence model to effectively discern the
51 dynamic evolution patterns of different variables.

52 To address these challenges, we propose Mamba-Branching. Mamba [14, 8] is a novel network
53 architecture characterized by its computational complexity that scales linearly with sequence length.
54 This represents a significant improvement over the quadratic complexity associated with Transform-
55 ers [43], making Mamba particularly well-suited for addressing challenge (1). Meanwhile, inspired
56 by CLIP [38], we employ contrastive learning to train the embedding layer prior to the overall
57 imitation learning process, effectively tackling challenge (2). Experimental results demonstrate that
58 Mamba-Branching outperforms all neural branching baselines across all real-world instances and
59 achieves superior solving efficiency over the advanced open-source solver SCIP’s default branching
60 rule on challenging instances.

61 2 Related Work

62 Learning-based approaches for accelerating MILP solving can be mainly divided into two
63 paradigms [5, 39]: replacing heuristic rules with neural networks within exact solution frame-
64 works and employing neural networks as primal heuristics. Research under the first paradigm
65 includes addressing branch variable selection [12, 15, 16, 40, 25, 26] and node selection [19, 27, 49]
66 problems within the B&B framework, as well as tackling cut selection issues in cutting-plane al-
67 gorithms [42, 22, 45]. These methods solely employ neural networks to replace heuristic rules
68 within exact solution frameworks, without compromising solution exactness. The second paradigm
69 aims to efficiently produce high-quality feasible solutions—rather than exact solutions—to tighten
70 the primal bound early in the process. A high-quality primal bound enables the B&B to eliminate
71 a significant number of non-promising nodes at an early stage through its pruning process. This
72 typically involves two key aspects: solution prediction [9, 36, 24, 18, 21, 33] and neighborhood
73 selection [46, 41, 20, 47]. The solution prediction approach typically employs neural networks to
74 predict optimal solutions, then uses these predictions to guide the search process. Neighborhood
75 selection starts from a feasible solution and fixes a subset of integer variables while optimizing the
76 remainder, with neural networks selecting which variables to fix.

77 Our work focuses on the generalization of neural branching variable selection policies, particularly
78 their ability to handle heterogeneous MILPs different from training instances. These approaches
79 can be mainly divided into two categories: parameterizing the B&B tree and diversifying training
80 instances. The first category aims to learn branching policies within a shared feature space across
81 different MILP instances. TreeGate [48] processes instance-independent features through a special-
82 ized neural architecture designed for branching decisions. Building on this, T-BranT [31] retains
83 historical data, modeling it as a graph structure processed by Graph Attention Networks for current
84 decision-making. The second category focuses on generating diverse instances and incorporating
85 them into the training of branching policies to enhance their generalization. AdaSolver [32] intro-
86 duces adversarial instance augmentation, which generates more diverse instances in directions that
87 hinder policy training. Meanwhile, MILP-Evolve [30] proposes a novel LLM-based evolutionary
88 framework capable of generating a large set of diverse MILP classes with an unlimited number of
89 instances. Specifically, our method falls into the first category. Using instance-independent features
90 as input, we also incorporate the sequential nature of the B&B tree into the decision-making process.

91 3 Preliminaries

92 3.1 B&B Algorithm and Branching Rules

93 **B&B Algorithm.** The standard form of MILPs is: $\arg \min_{\mathbf{x}} \{ \mathbf{c}^\top \mathbf{x} \mid \mathbf{A}\mathbf{x} < \mathbf{b}, \mathbf{x} \in \mathbb{Z}^p \times \mathbb{R}^{n-p} \}$,
 94 where the vector $\mathbf{x}$ represents n variables to be optimized, with p being the number of integer variables.
 95 $\mathbf{A}$, $\mathbf{b}$, $\mathbf{c}$ represent constraint matrix, constraint right term, and objective coefficient. For MILPs, an
 96 exact solution framework commonly used is B&B. This method first ignores the integer constraints
 97 to obtain and solve the relaxed problem at the root node. Subsequently, it iteratively searches for the
 98 global optimal solution through branching, bounding, and pruning. Branching involves selecting a
 99 variable with a fractional solution $x_i = b_i$ at the current node and adding the constraints $x_i < \lfloor b_i \rfloor$
 100 and $x_i \geq \lfloor b_i \rfloor + 1$ to form two child nodes. During bounding, the global upper and lower bounds
 101 (also known as the primal and dual bounds) are determined based on all existing nodes. The pruning
 102 process eliminates obviously infeasible nodes according to these bounds. This procedure repeats
 103 until the upper and lower bounds converge, yielding the global optimal solution.

104 **Branching Rules.** Here, the branching variable selection during B&B significantly impacts solving
 105 efficiency by influencing tree size. In [2], several heuristic branching rules are introduced. Among
 106 them, strong branching evaluates candidate variables by creating child nodes for each candidate and
 107 selecting the one maximizing dual bound improvement. While highly effective, this approach incurs
 108 significant computational overhead that counteracts its benefits for solution speed. Pscost branching
 109 guides current branching by leveraging historical branching records, avoiding extra computation but
 110 performing poorly early in the search tree due to insufficient branching records. Hybrid approaches
 111 combine both methods' advantages by using strong branching initially to establish reliable branching
 112 patterns, then transitioning to pscost branching once sufficient historical data is accumulated. The
 113 state-of-the-art (SOTA) relpscost branching [1], SCIP's default rule, implements this strategy—a
 114 variable's pscost is only considered trustworthy after undergoing sufficient strong branching steps.

115 3.2 Parameterized B&B Tree

116 To parameterize the B&B tree and obtain instance-independent features for heterogeneous MILP
 117 problems, Zarpellon et al. [48] design a state representation $s_t = (C_t, Tree_t)$. Here, $C_t \in \mathbb{R}^{|C_t| \times 25}$
 118 denotes candidate variable features, and $Tree_t \in \mathbb{R}^{o_1}$ represents tree features, where C_t denotes
 119 candidate variable set and $|C_t|$ represents candidate variable number. Since all features reflect the
 120 dynamic process of B&B trees, all MILP instances can be processed uniformly in the same feature
 121 space by a neural network named TreeGate, which jointly processes candidate and tree features
 122 through two components: a candidate network and a tree network. The candidate network first
 123 embeds each variable's 25-dimensional features into an h -dimensional space, then progressively
 124 reduces the dimensionality from h to d through multiple layers that halve the dimension at each step.
 125 Meanwhile, the tree network projects the tree features $Tree_t$ into an H -dimensional space (where
 126 $H = h + h/2 + \dots + d$) using a sigmoid activation to produce a gating vector $g \in [0, 1]^H$. This
 127 gating vector modulates the candidate network's layer outputs through element-wise multiplication.
 128 The final output $e_t \in \mathbb{R}^{|C_t| \times d}$ undergoes average pooling across the d -dimensional features, then
 129 being processed by a softmax layer to generate the candidate variable selection probabilities.

130 4 Methodology

131 In this section, we formally introduce Mamba-Branching, a neural branching policy specifically
 132 designed to capture the sequential structure of B&B trees. We begin by discussing the contrastive
 133 learning approach utilized for the embedding layer and the detailed design of the sequence inputs,
 134 followed by the detailed implementation of imitation learning. The overall framework of Mamba-
 135 Branching is illustrated in Figure 1.

136 4.1 Contrastive Learning for Embedding Layer

137 The embedding layer serves as a critical interface between raw state representations and downstream
 138 sequence models. In natural language processing (NLP), the success of embedding techniques
 139 has been well-established. These methods leverage the inherent distinguishability of discrete word
 140 tokens, where each word's unique identity naturally translates to separable embedding vectors through

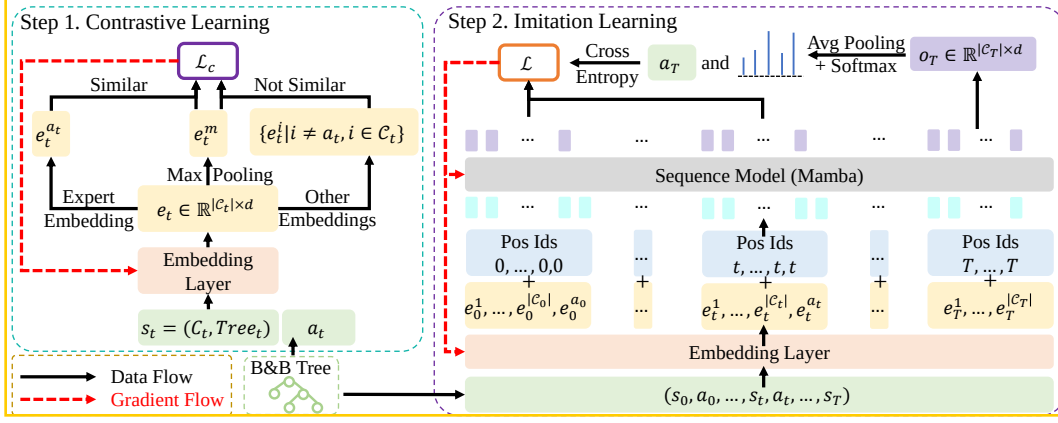


Figure 1: Overall framework of Mamba-Branching. The training process involves two stages: contrastive learning and autoregressive imitation learning. During the contrastive learning process, the state s_t and expert decision a_t at each branching step t are used to train the embedding layer via the designed contrastive loss function $\mathcal{L}_c$. During imitation learning, the branching trajectory $(s_0, a_0, \dots, s_T)$ is mapped to embeddings. At step t , expanded variable embeddings $(e_t^1, \dots, e_t^{|\mathcal{C}_t|})$ and expert embedding e_t^a form a group with shared positional encoding. These groups create a “branching path” input to the sequence model, where only outputs o_t corresponding to the variable embeddings are selected, with a_t serving as the label for imitation learning.

standard training paradigms. [4, 35] However, the branching variable selection problem in B&B presents a fundamentally different challenge. The state representation at each branching step t , denoted as $s_t = (C_t, Tree_t)$ (see subsection 3.2), contains a set of candidate variables C_t that frequently exhibit remarkably similar feature characteristics. This high degree of intra-step similarity arises from the shared constraints and problem structure inherent in combinatorial optimization problems. Unlike the clear distinctions between words in NLP tasks, the subtle but decision-critical differences between candidate variables in B&B require a more sophisticated approach to embedding learning.

To address this challenge, we develop a principled framework for learning discriminative embeddings in B&B decision making. The core of our approach lies in recognizing that effective branching decisions require the embedding space to maintain consistent separation between selected and non-selected variables. We formalize this requirement through Proposition 1. This condition specifies that the similarity between the selected variable’s embedding and a reference vector must exceed all other candidate similarities by a positive margin δ .

Proposition 1. For effective branching decisions, the embedding space must satisfy:

$$\forall t, \exists \delta > 0 \text{ s.t. } \text{sim}(e_t^a, e_t^m) \geq \max_{i \neq a} \text{sim}(e_t^i, e_t^m) + \delta, \quad (1)$$

where $\text{sim}(\cdot)$ is a similarity judgment function, e_t^i denotes embeddings, e_t^m is an anchor, a denotes the selected variable index.

To this end, before joint training, we first employ contrastive learning to train the embedding layer, enhancing its ability to differentiate between distinct candidate variable features. The loss function of contrastive learning is defined as $\mathcal{L}_c$, with the specific form as follows:

$$\mathcal{L}_c(\gamma) = \frac{1}{T} \sum_t \left(-\frac{e_t^m \cdot e_t^a}{\|e_t^m\| \|e_t^a\|} + \frac{1}{|\mathcal{C}_t| - 1} \sum_{i \neq a} \frac{e_t^m \cdot e_t^i}{\|e_t^m\| \|e_t^i\|} \right), \quad (2)$$

where γ denotes the parameters of TreeGate, T denotes the total number of branching steps, $e_t = \text{TreeGate}_\gamma(s_t)$, $e_t^m \in \mathbb{R}^d$ is the result of applying max-pooling to e_t along the $|\mathcal{C}_t|$ dimension, $e_t^a \in \mathbb{R}^d$.

The intuition behind this loss function design is to make the selected branching variable the most prominent and distinctive among all candidate variables. The max-pooling operation extracts a salient global feature as an anchor. By increasing the cosine similarity between the anchor and the selected

branching variable while decreasing the cosine similarity between the anchor and other candidate variables, the loss amplifies their differences and drives the feature of the selected branching variable toward the globally most salient direction.

4.2 Sequential Modeling Design

In B&B tree, nodes are progressively expanded until the upper and lower bounds converge. This process can be viewed as navigating through a complex maze to find a “branching path” from the root node to the optimal solution node. Traditional neural approaches to branching decisions have predominantly relied on the immediate state of the tree, neglecting the historical sequence of visited nodes and prior branching choices. This myopic perspective is fundamentally limiting, as it fails to leverage the rich sequential information inherent in the branching process. Just as an effective maze-solving strategy requires reasoning about the entire traversed path to avoid dead ends and redundant exploration, optimal branching decisions demand a holistic understanding of the search trajectory. This underscores the imperative for a paradigm shift toward path-aware sequential modeling.

To effectively model branching decisions, the sequence model must capture not only the sequential progression of states but also the intricate interrelationships among candidate variables within each state. Therefore, we explicitly encode the features of each candidate variable at all branching steps. We formally define the branching path $\mathbf{S}$ as a structured sequence of embeddings, where each state at step t is decomposed into its constituent candidate variables along with the selected branching decision. Specifically, $\mathbf{S}$ is represented as:

$$\mathbf{S} = [\underbrace{e_0^1, \dots, e_0^{|C_0|}, e_0^{a_0}}_{|C_0|+1}, \dots, \underbrace{e_t^1, \dots, e_t^{|C_t|}, e_t^{a_t}}_{|C_t|+1}, \dots, \underbrace{e_T^1, \dots, e_T^{|C_T|}}_{|C_T|}], \quad (3)$$

where $|C_t|$ denotes the number of candidate variables at branching step t , a_t represents the index of the selected branching variable, e_t denotes the embedding feature, and T is the maximum number of branching steps in the branching path. This formulation ensures that both the sequential dynamics and the variable-level interactions are preserved, enabling the model to leverage granular features for improved decision-making.

To ensure temporal coherence across branching steps, we employ positional encodings that assign identical positional indices to embeddings within the same step. The complete input representation $\mathbf{S}'$ is constructed by combining the branching path $\mathbf{S}$ with a learnable positional encoding matrix $\mathbf{E}_{pos}$:

$$\begin{aligned} pos &= [\underbrace{0, \dots, 0}_{|C_0|+1}, \dots, \underbrace{T, \dots, T}_{|C_T|}], \\ \mathbf{S}' &= \mathbf{E}_{pos} \oplus \mathbf{S}, \end{aligned} \quad (4)$$

where $\oplus$ denotes element-wise addition, $\mathbf{E}_{pos} \in \mathbb{R}^{(\sum_{t=0}^T |C_t| + T) \times d}$ represents the learnable positional encoding matrix obtained by mapping pos .

Subsequently, $\mathbf{S}'$ is fed into Mamba to obtain the output $\mathbf{O}_t$:

$$\begin{aligned} \mathbf{O}_t &= \text{Mamba}(\mathbf{S}') \\ &= [o_0^1, \dots, o_0^{|C_0|}, o_0^{a_0}, \dots, o_t^1, \dots, o_t^{|C_t|}, o_t^{a_t}, \dots, o_T^1, \dots, o_T^{|C_T|}], \end{aligned} \quad (5)$$

within each group, only the outputs corresponding to $|C_t|$ variable positions are extracted, denoted as $o_t = (o_t^1, \dots, o_t^{|C_t|})$, which are then processed through average pooling and softmax to obtain the variable probability distribution.

It can be observed that for the branching path $\mathbf{S}$, the sequence model actually needs to process an input length of $\sum_{t=0}^T |C_t| + T$. When either T or $|C_t|$ becomes large, the length of $\mathbf{S}$ increases substantially, presenting significant challenges to the sequence model’s ability to handle long sequences. Therefore, in addition to employing the most commonly-used Transformer Decoder as our sequence model, we also utilize Mamba [14] (see Appendix A for architectural details). In contrast to the Transformer’s quadratic complexity, Mamba achieves linear complexity relative to sequence length, making it unequivocally better suited for such long-sequence application scenarios. This is particularly critical in our application, where computational speed is paramount. If the neural branching policy’s inference complexity becomes excessively high and computationally prohibitive, it would fundamentally undermine our original objective of acceleration.

4.3 Imitation Learning under Autoregressive Paradigm

Following prior works [48, 31], we employ relpscost branching as the expert to collect demonstration datasets for imitation learning. In contrast to the commonly employed strong branching expert [12, 15], which are rarely applied in practical scenarios, relpscost provides a more realistic expert representation. For dataset collection, each instance is solved using SCIP. We sequentially record every state in the instance’s tree along with the corresponding relpscost-selected branching decisions, resulting in a complete trajectory denoted as $(s_0, a_0, s_1, a_1, \dots)$. In dataset $\mathcal{D}$, each instance’s trajectory is partitioned into fixed-length sub-trajectories for storage.

In Mamba-Branching, the branching policy is defined as π_θ , which operates in an autoregressive paradigm. The joint loss function $\mathcal{L}(\theta, \gamma)$ of embedding layer and sequence model is as follows:

$$\mathcal{L}(\theta, \gamma) = -\frac{1}{|\mathcal{D}|} \sum_{\tau \in \mathcal{D}} \sum_{(s_t, a_t) \in \tau} \log \pi_\theta(a_t | \tau_{0:t}), \quad (6)$$

where τ denotes a trajectory in $\mathcal{D}$, $\tau_{0:t} = (s_0, a_0, \dots, a_{t-1}, s_t)$, and $|\mathcal{D}|$ represents the total number of trajectories in $\mathcal{D}$. During inference, the predictions from previous branching steps serve as input for the current step, yielding the probability distribution $\pi_\theta(\cdot | \hat{\tau}_{0:t})$ over candidate variables, where $\hat{\tau}_{0:t} = (s_0, \hat{a}_0, \dots, \hat{a}_{t-1}, s_t)$.

5 Experiments

5.1 Setup

5.1.1 Benchmarks

MILP dataset. Our method is designed to maintain generalization capability across heterogeneous MILPs. Therefore, the training and test instances are deliberately constructed to be distinct, with the strict requirement that the test set should not contain any instances present in the training set. Following the selection of instances from previous works [48, 31], we construct two MILP datasets of different scales using instances from MIPLIB [13] and CORAL [29]: a smaller-scale dataset (MILP-S) and a larger-scale dataset (MILP-L). The MILP-S is entirely derived from [48], comprising 19 training instances and 8 test instances. MILP-L is constructed by expanding the dataset used in [31], containing 25 training instances and 73 test instances. For MILP-L’s test instances, we employ SCIP as the reference solver, categorizing 57 instances with solution times under 20 minutes as "easy" and 16 instances exceeding 20 minutes as "difficult". The details of MILP-S and MILP-L are provided in Appendix B.

Branching Dataset Collection. During data collection, consistent with previous works [48, 31], we employ random branching for the first r steps to enhance B&B exploration. After these r random steps, we switch to relpscost branching and collect the corresponding data. For each training instance, we configure $r \in \{0, 1, 5, 10, 15\}$ and collect training set using solver seeds $\{0, 1, 2, 3\}$, while reserving seed 4 exclusively for validation set.

5.1.2 Metrics

Nodes and Fair Nodes. The number of nodes in the B&B tree serves as a crucial metric for evaluating branching policies, as it directly impacts overall solving time. However, as noted in [10], this metric may be confounded by side effects of some sophisticated branching rules, such as strong branching. We therefore additionally employ the fair node number [10], which eliminates the confounding effects of these rules, thereby providing a more accurate reflection of the true capability of a branching policy. For branching policies that do not use strong branching, the number of nodes and fair nodes remains identical.

Primal-Dual Integral. For some challenging instances in MILP-L, obtaining optimal solutions may be computationally prohibitive, so a one-hour time limit is imposed. Under this constraint, node number becomes an inadequate metric for evaluating the performance of branching policies. In such cases, the primal-dual integral (PD integral) serves as a more appropriate evaluation criterion [11]. With a time limit T , the PD integral is expressed as $\int_0^T c^\top x_t^* - y_t^* dt$, where y_t^* is the best dual bound at time t , x_t^* is the best feasible solution at time t .

5.1.3 Baselines

We select two categories of branching policies as baselines: neural-based approaches and heuristic rules. The neural branching policies include: GCNN [12], TreeGate [48], T-BranT [31], and Transformer-Branching. GCNN is the most classical method and does not incorporate specific designs for heterogeneous MILPs. TreeGate and T-BranT are also based on instance-independent inputs, serving as the primary baselines for comparison with Mamba-Branching. Transformer-Branching employs Transformer as the sequence model to highlight Mamba’s advantages. The heuristic rules include random, pscost, and relpscost, where random and pscost represent the lower bounds of performance, relpscost serves as the expert and constitutes the upper bound of branching performance. However, neural branching policies may surpass relpscost in solving efficiency. More detailed reasons for the selection of baselines can be found in Appendix C.

5.1.4 Solver and Neural Policy Settings

Solver Settings. In our evaluation, we replace SCIP solver’s (v8.0.4) branching policy with our neural branching policy. To isolate the study of branching policies and eliminate interference from other solver components, we disable all primal heuristics and provide each test instance with a known optimal solution value as a cutoff. However, during branching data collection, we intentionally omit the cutoff to obtain longer branching sequences.

Neural Policy Settings. During Mamba-Branching training, the maximum branching step is $T = 99$, but as shown in subsection 4.2, its corresponding actual input length is considerably long. When using the Transformer as the sequence model, this length causes excessive GPU memory consumption that exceeds hardware limitations, thus $T = 9$ is adopted during Transformer-Branching training. For evaluation consistency, we employ autoregressive generation with $T = 24$ across all models. Additional implementation details and hyperparameters can be found in Appendix D.

Table 1: The experimental results on MILP-S. For the 8 test instances in MILP-S, each instance is evaluated with five random seeds {0,1,2,3,4} under a 1-hour time limit, and the results are presented as geometric means. Among them, blue background indicates the best results, bold font indicates the best results in neural policies, and $\star$ denotes reaching the time limit.

Method	Mamba-Branching	TreeGate	Transformer-Branching	T-BranT	GCNN	random	pscost	relpscost
Nodes	2054.99	2171.31	3078.56	2668.62	33713.63 $\star$	61828.29 $\star$	4674.34	730.21
Fair Nodes	2077.55	2205.06	3120.04	2715.16	33713.63 $\star$	61828.29 $\star$	4674.34	1227.25

5.2 Branching Performance

5.2.1 MILP-S

The experimental results in MILP-S can be found in Table 1, with the fair node results of all neural branching policies per instance shown in Figure 2. The single-step inference time comparison between Transformer and Mamba is shown in Figure 3. Notably, T-BranT necessitates at least one set of historical data, prompting the use of relpscost at the root node. This precise branching decision at the root significantly influences overall performance. For the sake of consistency, Mamba-Branching, TreeGate, and Transformer-Branching also employ relpscost at the root node, with Mamba-Branching and Transformer-Branching further leveraging it to initialize their input sequences. To evaluate the performance of pure neural branching, we additionally test variants that do not utilize relpscost initialization: TreeGate-p, Mamba-Branching-p, and Transformer-Branching-p, as shown in Table 3.

It can be observed that Mamba-Branching is the best branching policy besides relpscost. First, Mamba-Branching significantly outperforms the three lower-bound references: GCNN, random, and pscost. Compared with several neural branching policy baselines, whether initialized with relpscost or purely neural-based, Mamba-Branching surpasses T-BranT, TreeGate, and Transformer-Branching, achieving a new SOTA for neural branching policies. Additionally, in terms of single-step inference time, Mamba significantly outperforms Transformer, highlighting its advantage as a sequence model.

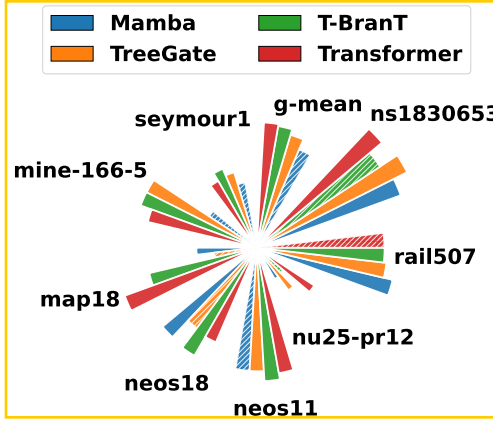


Figure 2: The fair node results of all neural branching policies in MILP-S.

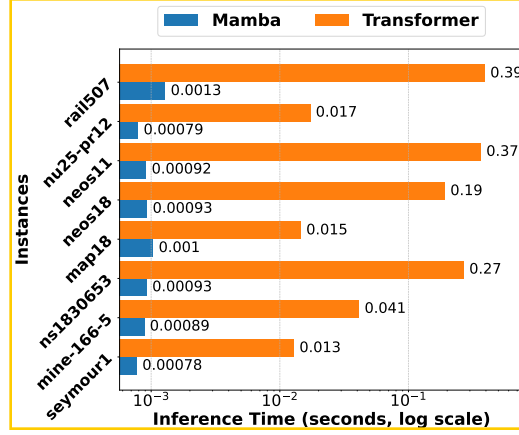


Figure 3: The inference time comparison between Mamba and Transformer in MILP-S.

5.2.2 MILP-L

In MILP-L, we further evaluate several methods that demonstrated strong performance in MILP-S, including TreeGate, T-BranT, Mamba-Branching, and relpscost. For the 57 easy instances, performance is assessed using both nodes and fair nodes. In contrast, for the 16 difficult instances, we use the PD integral with a 1-hour time limit as the evaluation metric. The results are presented in Table 2.

The results demonstrate that for easy MILPs, despite the expanded test instances compared to MILP-S, Mamba-Branching remains the best neural branching policy.

but still inferior to relpscost. For difficult instances, Mamba-Branching achieves the best PD integral performance among all branching policies, even surpassing relpscost. This indicates that within the same time limit, Mamba-Branching enables the fastest convergence of primal and dual bounds.

Table 2: The experimental results on MILP-L. Consistent with the experiments on MILP-S, all instances are evaluated with 5 random seeds under a 1-hour time limit, with results reported as geometric means. Blue background indicates the best results and **bold** font indicates the best results in neural policies.

Method	Easy		Difficult
	Nodes	Fair Nodes	PD Integral
Mamba-Branching	1819.32	2053.91	12319.55
TreeGate	2218.29	2534.09	14625.68
T-BranT	2009.77	2298.76	13538.47
relpscost	667.93	1455.49	12741.36

5.2.3 Discussion

Advantage of Sequential Nature. First, Mamba-Branching consistently outperforms TreeGate and T-BranT across all scenarios due to its consideration of the sequential nature of B&B trees. Neither TreeGate (which completely ignores historical data) nor T-BranT (which utilizes historical data non-sequentially) achieves the effectiveness of sequential historical data utilization. This aligns with our maze analogy in subsection 4.2: sequentially recalling paths facilitates better current decision-making.

Limitation of Transformer. Transformer-Branching also leverages sequential nature but performs poorly, with Transformer-Branching-p even underperforming the lower-bound pscost. The suboptimal performance stems from its 10-step branching history limit during training (due to hardware constraint), while Mamba-Branching accommodates 100 steps. Furthermore, the inference time comparison in Figure 3 demonstrates that in our time-sensitive scenario aimed at reducing solving time, Transformer is entirely unsuitable as a branching policy. The underlying reason here is that Transformer’s complexity is quadratic with respect to sequence length, while Mamba’s is linear. Although Transformer is theoretically suitable as a sequence model, employing Mamba offers greater practicality and feasibility. A more detailed comparison can be found in Appendix E.

Factors Outperforming Relpscost. As for relpscost, Mamba-Branching does not outperform in easy instances but surpasses it in difficult ones. The reason can be summarized as follows: (1) Relpscost is a hybrid method combining strong and pscost branching, incorporating a reliability criterion: a

Table 3: On MILP-S, the results of pure neural branching policies TreeGate-p, Transformer-Branching-p, and Mamba-Branching-p, as well as Mamba-Branching-p without contrastive learning (w/o cl). The experimental setup remains consistent with the aforementioned configuration on MILP-S.

Method	Mamba-Branching-p	Mamba-Branching-p (w/o cl)	Transformer-Branching-p	TreeGate-p
Nodes	2272.43	3000.92	5138.15	3179.55
Fair Nodes	2272.43	3000.92	5138.15	3179.55

variable can only switch to pscost after being selected by strong branching a certain number of times. Therefore, for difficult instances with more variables, the initialization process is time-consuming, leading to potential inefficiency. In contrast, neural policies benefit from fast inference and exhibit advantages on difficult instances. (2) In relpscost, the use of pscost for leveraging historical data does not account for the sequential nature, whereas Mamba-Branching explicitly incorporates this consideration. (3) As mentioned in [48], the relpscost in SCIP has been fine-tuned for a large number of instances, resulting in excellent performance on easy instances. However, for more complex and challenging instances, such parameter tuning may not provide adequate coverage.

5.3 Ablation Study

In this section, an ablation experiment is conducted to verify the role of contrastive learning. First, the most straightforward comparison is to evaluate the branching performance difference when contrastive learning is applied or not to the embedding layer. Under pure neural branching, the results on MILP-S without and with contrastive learning are denoted as Mamba-Branching-p (w/o cl) and Mamba-Branching-p, respectively, as shown in Table 3. Meanwhile, to demonstrate that contrastive learning indeed achieves its intended effect, that is, making the feature of expert-selected variable more distinguishable compared to other candidates, we also visualize the t-SNE-reduced [34] embeddings, as shown in Figure 4.

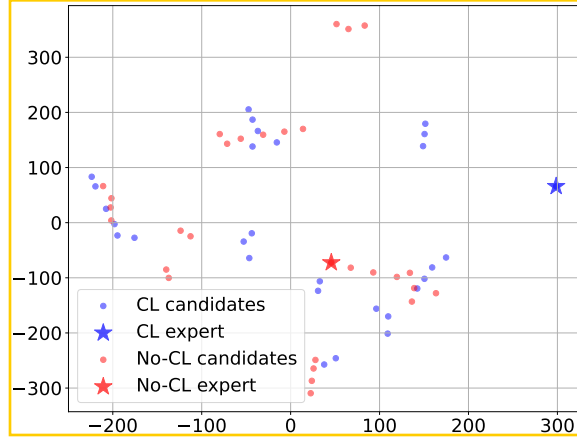


Figure 4: At a random given state, the embeddings with and without contrastive learning.

First, in terms of branching performance, Mamba-Branching-p demonstrates superior results compared to its counterpart without contrastive learning, Mamba-Branching-p (w/o cl). The visualization then reveals that with contrastive learning, the expert-selected variable exhibits greater outlier characteristics and enhanced discriminability relative to other candidate variables. In contrast, without contrastive learning, the expert-selected variable becomes less distinguishable and tends to cluster near candidate variables.

6 Conclusion and Future Work

In this paper, we propose Mamba-Branching, the first approach to consider the sequential nature in B&B trees. To address the challenges of long sequences and embedding distinctiveness posed by sequential nature, we employ Mamba as the sequence model and design a contrastive learning method to train the embedding layer, enabling the sequence model to distinguish between different candidate variables. In experiments, Mamba-Branching outperforms all neural branching policies and achieves superior solving efficiency compared to relpscost on challenging instances. One limitation of our approach is the reliance on imitation learning, which requires a time-consuming collection of expert demonstrations. In future work, we will focus on investigating the potential of sequential nature in reinforcement learning-based branching policies, thereby eliminating the dependency on expert data.

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554 A Mamba Architecture

555 Mamba is a novel network architecture based on State Space Model (SSM) that may potentially
 556 replace self-attention-based Transformer models [14, 8, 44]. SSM is a concept originating from
 557 control theory, with its earliest roots traceable to the classical Kalman filter [23]. The continuous-time
 558 formulation of SSM can be represented as follows:

$$\begin{aligned} \dot{\mathbf{z}}(t) &= \mathbf{M}(t)\mathbf{z}(t) + \mathbf{N}(t)\mathbf{u}(t) \\ \mathbf{y}(t) &= \mathbf{P}(t)\mathbf{z}(t), \end{aligned} \quad (7)$$

559 where $\mathbf{z}(t) \in \mathbb{R}^z$, $\mathbf{y}(t) \in \mathbb{R}^q$, $\mathbf{u}(t) \in \mathbb{R}^u$. After zero-order hold discretization, the discrete-time
 560 formulation is obtained as follows:

$$\begin{aligned} \mathbf{z}_t &= \bar{\mathbf{M}}\mathbf{z}_{t-1} + \bar{\mathbf{N}}\mathbf{u}_t \\ \mathbf{y}_t &= \mathbf{P}\mathbf{z}_t, \end{aligned} \quad (8)$$

561 where $\bar{\mathbf{M}} = \exp(\Delta\mathbf{M})$, $\bar{\mathbf{N}} = (\Delta\mathbf{M})^{-1}(\exp(\Delta\mathbf{M}) - \mathbf{I}) \cdot \Delta\mathbf{N}$, Δ denotes the step size.

562 To meet the parallelization requirements of the training process, the SSM can alternatively be
 563 represented as follows:

$$\begin{aligned} \bar{\mathbf{K}} &= (\mathbf{P}\bar{\mathbf{N}}, \mathbf{P}\bar{\mathbf{M}}\bar{\mathbf{N}}, \dots, \mathbf{P}\bar{\mathbf{M}}^k\bar{\mathbf{N}}) \\ \mathbf{y} &= \bar{\mathbf{K}} * \mathbf{u}. \end{aligned} \quad (9)$$

564 Mamba builds upon SSM by introducing Selective SSM, which essentially treats $\bar{\mathbf{N}}$, $\mathbf{P}$ and Δ as
 565 functions of the input while keeping $\bar{\mathbf{M}}$ unchanged. From a control theory perspective, this transforms
 566 the system from time-invariant to time-varying. Furthermore, Mamba incorporates hardware-aware
 567 algorithm design that enables efficient storage of intermediate results through parallel scanning,
 568 kernel fusion, and recalculation.

569 B Benchmark Details

570 All training and test instances in MILP-S are listed in Table 6. The training instances in MILP-L
 571 are presented in Table 7. All easy test instances in MILP-L are shown in Table 8, while all difficult
 572 test instances are presented in Table 9. Here, an instance is considered easy if SCIP’s solving time
 573 is less than 20 minutes; otherwise, it is classified as difficult. These instances are sourced from
 574 MIPLIB [13] and CORAL [29], all collected from real-world application scenarios, with specific
 575 instance selections referenced to [48] and [31]. Serving as benchmarks, these instances effectively
 576 reflect the practical significance of neural branching policies in real-world applications.

577 C Detailed reasons for Baseline Selection

578 The neural branching policies and their selection rationale are detailed below: (1) GCNN [12]: This
 579 method is not designed for heterogeneous MILPs and performs poorly on unseen MILP instances
 580 outside the training distribution. The experimental results of GCNN highlight the advantage of
 581 instance-independent feature design in terms of generalization. (2) TreeGate [48]: The TreeGate
 582 network incorporates instance-independent inputs by design, making it suitable for heterogeneous
 583 MILPs. Additionally, since Mamba-Branching’s embedding layer adopts the TreeGate architecture,
 584 TreeGate serves as a critical control group for our method. (3) T-BranT [31]: Building upon Tree-
 585 Gate’s feature design, T-BranT employs attention to capture mutual connections among candidates.
 586 Meanwhile, T-BranT processes historical data from an unordered graph perspective. The comparison
 587 with T-BranT serves to evaluate whether Mamba-Branching’s sequential processing of historical
 588 data demonstrates superior performance over T-BranT’s unordered graph approach. (4) Transformer-
 589 Branching: When selecting a sequence model for our approach, the Transformer would naturally
 590 be the most immediate consideration. Thus, we include Transformer-Branching as a comparative
 591 baseline against Mamba-Branching, specifically to highlight the advantages of employing Mamba as
 592 the sequence model.

593 The heuristic rules are selected with the following rationale: (1) Random: Serves as the performance
 594 lower bound, demonstrating the detrimental effects of completely omitting a deliberate branching

policy. (2) Pscost: A purely historical data-driven branching method that, like Random, also establishes a performance lower bound. (3) Relpscost: The expert policy in the imitation learning of Mamba-Branching. Simultaneously, it is also the SOTA heuristic rule and the default rule in SCIP. Relpscost serves as the upper bound of decision accuracy for neural branching policies. However, benefiting from the fast inference speed of neural networks, neural branching policies may surpass relpscost in terms of efficiency.

D Implementation Details

All experiments in this paper are run on NVIDIA A100-PCIE-40GB GPU and Intel(R) Xeon(R) Gold 5218 CPU. The hyperparameters in the training of Mamba-Branching are shown in Table 4.

During training, sequences consisting of every 100 branching steps are fed as input to Mamba. Although the number of candidate variables varies across batches, several hundred candidates represent a typical scenario. Consequently, the actual input sequence length may extend to tens of thousands of tokens – a scale that would easily trigger GPU memory overflow if using Transformer architectures. We therefore adopt Mamba as our sequence model, whose computational complexity scales linearly with sequence length, to effectively circumvent these hardware limitations.

During inference, the sequence fed into Mamba consists of: (1) states and predicted actions from the most recent 24 branching steps, and (2) the current state. Unlike the training phase, we deliberately reduce the number of branching steps to maintain sufficiently short inference latency, thereby reducing the total solving time. After systematic tuning, we ultimately select 25 branching steps ($T_{eva}=24$) to achieve an optimal balance between branching accuracy and inference speed.

Table 4: The hyperparameters of Mamba-Branching

Name	Description	Value
d	Output dimension of the candidate net in TreeGate, which is equivalent to the embedding size of Mamba.	8
h	Hidden state dimension of the Candidate Net in TreeGate.	64
depth	Layer number of Tree Net in TreeGate.	3
batch_size	Batch size of Mamba Training	32
lr_cl	Learning rate of contrastive learning.	0.0001
optimizer_cl	Optimizer of contrastive learning.	Adam
lr	Learning rate of imitation learning.	0.001
optimizer	Optimizer of imitation learning.	AdamW
wd	Weight decay coefficient of imitation learning.	0.01
T_train	Maximum branching steps considered during training.	99
T_eva	Maximum branching steps considered during evaluating.	24
d_state	SSM state expansion factor in Mamba	64
d_conv	Local convolution width in Mamba	4
expand	Block expansion factor in Mamba	2

E Computational Complexity Comparison between Transformer and Mamba

As is well-known, Mamba exhibits linear complexity with respect to sequence length, while Transformer demonstrates quadratic complexity. In this section, we present experimental results that provide a detailed comparison of the complexity between Mamba and Transformer when employed as branching policies. Our complexity analysis focuses on two key aspects: space complexity and time complexity. For space complexity, we compare the GPU memory consumption of Mamba and Transformer during both training and inference phases. Regarding time complexity, we primarily examine the inference latency of both models when functioning as branching policies. The experimental results are shown in Table 5.

As shown in the experiments, when processing 100 branching steps during training (even with a batch size of 1), Transformer-Branching fails to train altogether, while Mamba-Branching occupies minimal GPU memory. During inference, Mamba-Branching demonstrates significantly lower GPU

memory consumption and inference time. In contrast, Transformer-Branching not only requires substantially more GPU memory but, more critically, suffers from prohibitively long inference times. Since the fundamental purpose of adopting a neural branching policy is to accelerate MILP solving, such excessive inference time directly contradicts our original objective.

Table 5: A comparison of computational complexity between Mamba-Branching and Transformer-Branching. During training, we uniformly set $T_{\text{train}}=99$ with a batch size of 1. For inference, we consistently use $T_{\text{eval}}=24$. After collecting 25 Branching steps, we measure the network’s inference time and GPU memory consumption, take the geometric mean across all test instances of MILP-S.

Method	GPU memory of Train (GB)	GPU memory of Inference (GB)	Inference Time (s)
Mamba-Branching	0.017	0.013	0.00093
Transformer-Branching	out of memory	1.051	0.075

Table 6: All instances in MILP-S.

Instance	Variables	Constraints	Set
air04	8904	823	train
air05	7195	426	train
dcmulti	548	473	train
eil33-2	4516	32	train
istanbul-no-cutoff	5282	20346	train
l152lav	1989	97	train
lseu	89	28	train
misc03	160	96	train
neos20	1165	2446	train
neos21	614	1085	train
neos-476283	11915	10015	train
neos648910	814	1491	train
pp08aCUTS	240	246	train
rmatr100-p10	7359	7260	train
rmatr100-p5	8784	8685	train
sp150x300d	600	450	train
stein27	27	118	train
swath1	6805	884	train
vpm2	378	234	train
map18	164547	328818	test
mine-166-5	830	8429	test
neos11	1220	2706	test
neos18	3312	11402	test
ns1830653	1629	2932	test
nu25-pr12	5868	2313	test
rail507	63019	509	test
seymour1	1372	4944	test

Table 7: Training instances in MILP-L

Instance	Variables	Constraints
30n20b8	18380	576
air04	8904	823
air05	7195	426
cod105	1024	1024
comp21-2idx	10863	14038
demulti	548	290
eil33-2	4516	32
istanbul-no-cutoff	5282	20346
l152lav	1989	97
lseu	89	28
misc03	160	96
neos20	1165	2446
neos21	614	1085
neos-476283	814	1491
neos648910	11915	10015
pp08aCUTS	240	246
rmatr100-p10	8784	8685
rmatr100-p5	7359	7260
rmatr200-p5	37816	37617
roi5alpha10n8	106150	4665
sp150 $\times$ 300d	600	450
stein27	27	118
supportcase7	138844	6532
swath1	6805	884
vpm2	378	234

Table 8: Easy test instances in MILP-L , SCIP's solving time is less than 20 minutes.

Instance	Variables	Constraints	SCIP Solving Time
aflow40b	2728	1442	375.32
app1-2	26871	53467	662.56
bc1	1751	1913	237.68
bell3a	133	123	1.54
bell5	104	91	0.63
biella1	7328	1203	271.30
binkar10_1	2298	1026	47.79
blend2	353	274	0.37
dano3_5	13873	3202	189.77
fast0507	63009	507	150.11
map10	164547	328818	515.00
map18	164547	328818	250.49
map20	164547	328818	218.78
mik-250-20-75-4	270	195	55.67
mine-166-5	830	8429	36.83
misc07	260	212	28.94
n2seq36q	22480	2565	497.79
neos11	1220	2706	171.35
neos12	3983	8317	674.23
neos-1200887	234	633	16.46
neos-1215259	1601	1236	110.12
neos13	1827	20852	96.20
neos18	3312	11402	27.63
neos-4722843-widden	77723	113555	864.13
neos-4738912-atrato	6216	1947	304.32
neos-480878	534	1321	52.92
neos-504674	844	1344	114.11
neos-504815	674	1067	34.61
neos-512201	838	1337	42.39
neos-584851	445	661	7.11
neos-603073	1696	992	269.58
neos-612125	9554	1795	43.31
neos-612162	9893	1859	40.85
neos-662469	18235	1085	566.6
neos-686190	3660	3664	61.03
neos-801834	3220	3300	51.71
neos-803219	640	901	32.99
neos-807639	1030	1541	20.52
neos-820879	9522	361	56.56
neos-829552	40971	5153	353.99
neos-839859	1975	3251	64.10
neos-892255	1800	2137	54.07
neos-950242	5760	34224	149.60
ns1208400	2883	4289	111.91
ns1830653	1629	2932	148.27
nu25-pr12	5868	2313	22.01
nw04	87482	36	42.85
p0201	201	133	0.81
pg	2700	125	45.39
pp08a	240	136	1.44
rai507	63019	509	160.55
roll3000	1166	2295	44.27
rout	556	291	39.51
satellites1-25	9013	5996	952.04
seymour1	1372	4944	61.91
sp98ir	1680	1531	92.98
unitcal_7	25755	48939	889.85

Table 9: Difficult test instances in MILP-L , SCIP's solving time is more than 20 minutes.

Instance	Variables	Constraints	SCIP Solving Time
atlanta-ip	48738	21732	3600.00
bab5	21600	4964	2665.41
harp2	2993	112	1642.38
map16715-04	164547	328818	2423.74
msc98-ip	21143	15850	3600.00
mspp16	29280	561657	2722.23
n3seq24	119856	6044	3600.00
pigeon-10	490	931	3600.01
bab2	147912	17245	3600.02
bab6	114240	29904	3600.01
neos-4338804-snowy	1344	1701	3600.08
neos-4387871-tavua	4004	4554	3600.00
neos-4647030-tutaki	12600	8382	3600.09
nursesched-medium-hint03	34248	14062	3600.00
opm2-z10-s4	6250	160633	3600.01
radiationm40-10-02	172013	173603	3600.00

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